

Covered: Submodules, Torsion, module homomorphisms, isomorphisms, direct sums, cyclic submodules, generating sets, linear (in)dependence

1 Power Series Ring

Definition 1.1. Let R be a commutative ring. We call $\sum_{i=0}^{\infty} a_i x^i$ with coefficients in R a *formal power series* in one variable x .

Given

$$F = \sum_{i=0}^{\infty} a_i x^i, \quad G = \sum_{i=0}^{\infty} b_i x^i,$$

we define

$$F + G = \sum_{i=0}^{\infty} (a_i + b_i) x^i, \quad FG = \sum_{i=0}^{\infty} c_i x^i, \quad \text{where } c_i = \sum_{k=0}^i a_k b_{i-k}$$

Let $R[[x]]$ denote the set of all formal power series in x with coefficients in R . With the above operations, $R[[x]]$ is a commutative ring.

Given n variables $x_1, \dots, x_n$, define $R[[x_1, \dots, x_n]]$ to be the set of all formal power series of the form

$$\sum_{\mathbf{i} \in \mathbb{N}_0^n} a_{\mathbf{i}} x_1^{i_1} \cdots x_n^{i_n}, \quad a_{\mathbf{i}} \in R,$$

where $\mathbf{i} = (i_1, \dots, i_n)$ and $i_j \geq 0$. In fact,

$$R[[x_1, \dots, x_n]] \cong R[[x_1]][[x_2]] \cdots [[x_n]],$$

and, in particular,

$$R[[x, y]] \cong R[[x]][[y]].$$

Theorem 1.2. If R is Noetherian, then $R[[x]]$ is Noetherian. In fact,

$$R[[x_1, \dots, x_n]]$$

is Noetherian for every finite n .

Remark. If F is a field, then $F[[x]]$ is a UFD (indeed, every nonzero element can be written uniquely in the form $x^n u$, where u is a unit).

$$R = \mathbb{Z}[\sqrt{-5}], \quad I = (2, 1 + \sqrt{-5}) = 2R + (1 + \sqrt{-5})R.$$

Let

$$a = x_1 + y_1 \sqrt{-5}, \quad b = x_2 + y_2 \sqrt{-5}, \quad x_1, y_1, x_2, y_2 \in \mathbb{Z}.$$

Then

$$\begin{aligned} 2a + b(1 + \sqrt{-5}) &= 2(x_1 + y_1 \sqrt{-5}) + (x_2 + y_2 \sqrt{-5})(1 + \sqrt{-5}) \\ &= 2x_1 + 2y_1 \sqrt{-5} + x_2 + x_2 \sqrt{-5} + y_2 \sqrt{-5} - 5y_2 \\ &= (2x_1 + x_2 - 5y_2) + (2y_1 + x_2 + y_2) \sqrt{-5}. \end{aligned}$$

Thus every element of I is of the form

$$m + n \sqrt{-5}, \quad m, n \in \mathbb{Z},$$

subject to the corresponding congruence conditions coming from the coefficients above.

2 Modules

Definition 2.1. Let R be a commutative ring. An R -module M is an abelian group $(M, +)$ together with a scalar multiplication

$$R \times M \longrightarrow M, \quad (r, m) \longmapsto rm,$$

such that for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$,

$$1m = m \tag{1}$$

$$r(m_1 + m_2) = rm_1 + rm_2 \tag{2}$$

$$(r_1 + r_2)m = r_1m + r_2m \tag{3}$$

$$(r_1r_2)m = r_1(r_2m) \tag{4}$$

Thus the scalar multiplication is compatible with both the additive structure of M and the ring structure of R .

$$r_1(r_2m) = (r_1r_2)m.$$

Example 2.2. 1. If R is a field and $M = V$ is a vector space over R , then V is an R -module.

2. If $I \subseteq R$ is an ideal, then I is an R -module under the usual addition and scalar multiplication.

3. If $I \subseteq R$ is an ideal, then R/I is an R -module under

$$r \cdot (a + I) := ra + I.$$

4. Let $M = \{f : \mathbb{R} \rightarrow \mathbb{R}^3\}$. Under pointwise addition and scalar multiplication,

$$(f + g)(x) = f(x) + g(x), \quad (rf)(x) = rf(x),$$

M is an R -module (in particular, an $\mathbb{R}$ -module when $R = \mathbb{R}$).

5. Every abelian group is a $\mathbb{Z}$ -module, with

$$n \cdot m = \begin{cases} \underbrace{m + \cdots + m}_{n \text{ times}}, & n > 0, \\ 0, & n = 0, \\ -\underbrace{(m + \cdots + m)}_{|n| \text{ times}}, & n < 0. \end{cases}$$

Let M be an R -module. Suppose

$$am = 0 \quad \forall m \in M, \forall a \in I,$$

for some ideal $I \trianglelefteq R$. Then we say that M is *annihilated by I* . In that case M is naturally an R/I -module, with

$$(r + I)m := rm.$$

This is well-defined since if $r + I = s + I$, then $r - s \in I$, and hence

$$(r - s)m = 0.$$

Definition 2.3. Let M be an R -module and let $A = \{a_1, \dots, a_n\} \subseteq M$. An element

$$r_1a_1 + \cdots + r_na_n, \quad r_i \in R,$$

is called a *linear combination* of $a_1, \dots, a_n$.

Definition 2.4. We say that $\{m_1, \dots, m_n\} \subseteq M$ is a *spanning set* of M if every element of M is a linear combination of $m_1, \dots, m_n$. In this case,

$$M = \text{span}\{m_1, \dots, m_n\}.$$

Definition 2.5. An R -module M is called *finitely generated* if it has a finite spanning set.

Definition 2.6. Let M be an R -module. We say $N \subseteq M$ is a *submodule* if N is an R -module under the operations inherited from M .

Equivalently,

$$N \neq \emptyset, \quad a + b \in N \quad \forall a, b \in N, \quad ra \in N \quad \forall r \in R, a \in N.$$

3 Algebras

Definition 3.1. A ring A is an R -algebra if there exists a ring homomorphism $\varphi : R \rightarrow A$ such that

$$\varphi(1_R) = 1_A, \quad \varphi(R) \subseteq Z(A),$$

where $Z(A)$ denotes the centre of A .

Equivalently,

$$\varphi(r)a = a\varphi(r) \quad \forall a \in A, r \in R.$$

In this case A is naturally an R -module via

$$r \cdot a := \varphi(r)a.$$

Definition 3.2. Let A, B be R -algebras. An R -algebra homomorphism from A to B is a ring homomorphism $\psi : A \rightarrow B$ such that

$$\psi(1_A) = 1_B \tag{5}$$

$$\psi(\varphi_A(r)) = \varphi_B(r) \quad \forall r \in R. \tag{6}$$

Equivalently,

$$\psi(ra) = r\psi(a) \quad \forall r \in R, a \in A.$$